

Crossplay & multi-version (first-party Geyser + Via class)

YaPcore aims for **one shared world** with first-party protocol coverage:

- **Bedrock:** `GeyserStyleTranslator` / `CrossplayHub` — not the Geyser jar
- **Older / other JE:** `ProtocolCompat` / `ViaStyleRemapper` — not Via* jars
- **Floodgate-class auth:** core `FloodgateAuth` + backend `yap-floodgate.jar` behind Velocity/Link

Supported JE floor: 1.20.2+ onto Folia/Paper 26.2. Bedrock smoke: `geyserParitySmoke=true` on 1.21.50.

Product note: Default `game-authority=folia`. Phase 4 join/spawn + core play-depth are green. Wave 2 closes inventory/forms honesty for native Bedrock; Floodgate-only and some UIs are **Limited** / **Out** (not silent Partial).

Connection paths

Path	Transport	Forms	Action bar / sidebar	Inventory authority
Native YaPcore Bedrock	UDP dual-stack	Chassis <code>FormService</code> (simple/modal/custom)	<code>BedrockUiBridge</code>	Shadow + Paper inject
Velocity/Geyser → YaPFloodgate	JE TCP to Folia	Limited — no chassis session; chat explains forms need native UDP	Paper action bar + scoreboard	Paper Bukkit inventory
YaP Link + Bedrock on YaPcore gateway	Proxy UDP terminating on chassis	Same as native	Same	Same

Fidelity matrix (Wave 2)

ID	Area	Status
Join / spawn	JE + Bedrock smoke	Green
Dig / place / chat / commands	Shared world ops	Green
G.25	Entity health/nametag	Green
G.28	<code>/clear</code> inventory	Green

ID	Area	Status
G.31	Title / bossbar / scoreboard	Green
G.33	Placed skull block actor	Green (owner name)
G.33	Item-in-hand player heads	Green (SkullOwner Name NBT); full profile hash texture = Stretch
G.34	JE pack → BE offer	Green
P4.6	Chest / furnace / hopper open	Green (live Paper resync on push)
P4.6	Enchant / workbench / villager	Green (best-effort); XP/layout polish ongoing
Forms (native UDP)	Simple / modal / custom	Green
Forms (Floodgate-only)	MMO / admin forms	Limited — explicit user message; not Partial
Anvil / smithing / loom / stonecutter / cartography	Container UIs	Out — use Java Edition
Full Geyser feature matrix	Every BE packet	Out — YaP intentional depth

Streamlined one-port join

```

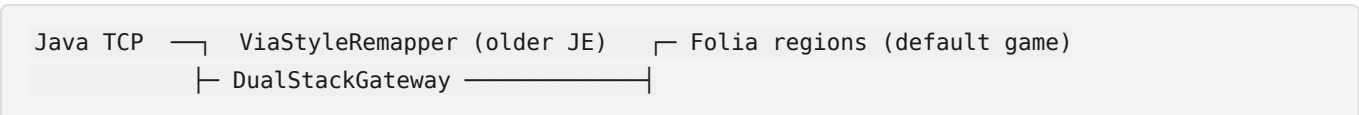
shared-listen-port=true
crossplay-enabled=true
port=25566
bedrock-port=25566

```

Edition	Socket (local bind)	Same-PC	Public (nginx / SRV)
Java	TCP :25566	127.0.0.1:25566	yapcoremc.yaplabs.us:25565
Bedrock	UDP :25566	127.0.0.1:25566	yapcoremc.yaplabs.us:25565

Disable with `shared-listen-port=false` for a separate Bedrock UDP port.

Architecture



```
Bedrock UDP └─ CrossplayHub ── UnifiedPlayer roster
               GeyserStyleTranslator
               FloodgateAuth (core) / yap-floodgate (Velocity / YaP Link)
```

Scope

Target: Geyser-class + Via-class coverage in YaP code for supported bands. **Not claimed:** stock Geyser jar parity, Floodgate-only forms, or every complex JE container on Bedrock.

GUI

- **Connect** — Crossplay address + Copy
- **Settings** — Shared listen port + Crossplay toggles
- **nginx** — domain / stream ports

Related

- [MMO_BEDROCK_UI.md](#) — forms need native session
- [VELOCITY.md](#) — Floodgate behind proxy
- [YAPCORE_WHITEPAPER.md](#)